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Professor Nils Thürey
Technical University of Munich (TUM)
Focus Group for Simulation and Digital Twin
Munich, Germany

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Subject: Application for Doctoral Research - Differentiable Physics & Digital Twins

Dear Professor Thürey,

I am writing to formally submit my application for a doctoral position within your Focus Group for Simulation and Digital Twin at the Technical University of Munich. As an AI Engineer with a deep interest in the synergy between deep learning and physical simulations, I have closely followed your work on differentiable physics and generative modeling for fluid dynamics.

Currently, I am finalizing my state engineering thesis at **Sonatrach**, where I have designed and implemented a **Generative Digital Twin** for the monitoring of gas turbine systems. My research addresses the challenge of making neural surrogates "physics-aware" by embedding conservation laws into the training process using **PINN** frameworks. I have found that this approach is critical for the reliability of digital twins in safety-critical industrial environments.

I am particularly inspired by your efforts to combine classical solvers with neural networks to create faster and more accurate simulation tools. My technical experience with **Graph Neural Networks (GNNs)** for modeling system topology and **Mamba-SSM** for high-performance sequential modeling allows me to approach physical systems from a multiscale perspective. I am eager to explore how these architectures can be used to develop **Differentiable Industrial Surrogates** that can be trained on a mix of simulation data and real-world sensor streams.

TUM is my first choice for a PhD because it is at the forefront of the Digital Twin revolution. I am a highly motivated researcher, proficient in Python and PyTorch, and I am eager to bring my industrial experience at Sonatrach to your research group.

I have attached my academic transcripts, my CV (noting ESI-SBA's H+ recognition), and a research proposal for your review. I would be honored to discuss how I can contribute to the state-of-the-art research in your laboratory.

Thank you for your time and consideration.

Sincerely,

Ismail AISSAOUI